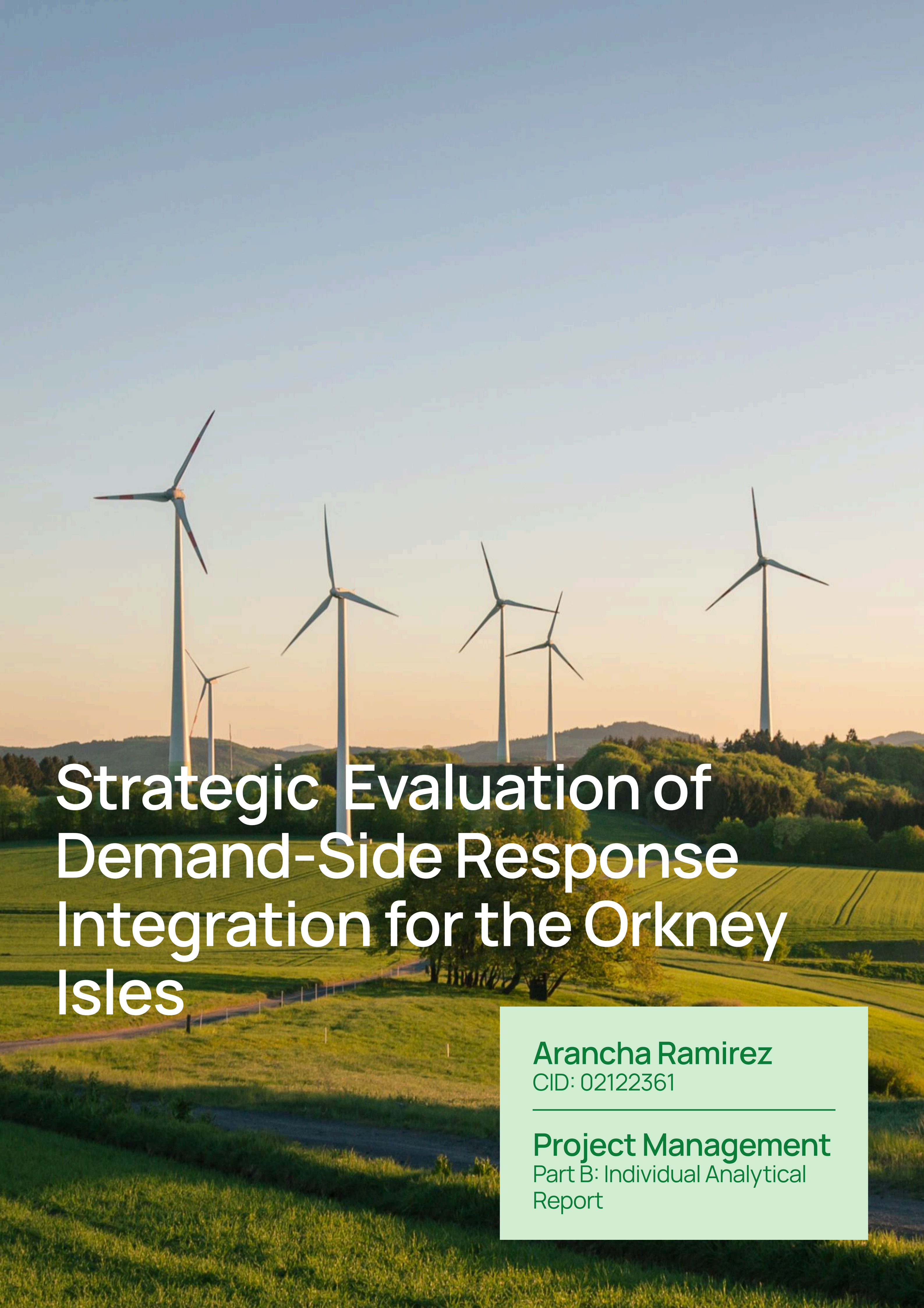

Strategic Evaluation of Demand-Side Response Integration for the Orkney Isles

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Project Management
Part B: Individual Analytical Report

Project Management

The Structural Paradox of the Orkney Energy Ecosystem

The Orkney Isles occupy a unique position within the global energy transition, serving as both a pioneering laboratory for renewable innovation and a stark example of curtailment that threatens decarbonization efforts. Historically, the archipelago has over-performed in renewable generation, currently producing approximately 115% of its total annual electricity demand through a diversified fleet of 23 large-scale wind turbines. [1] However, this achievement is frequently neutralized by a fundamental physical and regulatory constraint: the regional electricity grid cannot always absorb or export the total volume of energy produced. During periods of peak wind output and low local demand, or when the subsea transmission cables to the Scottish mainland reach their limit, the regional Active Network Management (ANM) system intervenes by triggering curtailment, reducing or shutting down these renewable assets. [2]

This curtailment represents a significant socioeconomic and technical inefficiency. While clean energy is being discarded, approximately 63% of Orkney households struggle with fuel poverty, a rate substantially higher than the Scottish national average. [3]

The proposed solution is an automated Demand-Side Response (DR) system that orchestrates smart appliances and boilers to synchronise residential consumption with peak wind generation, effectively transforming curtailed surplus energy into affordable heat for fuel-poor households.

Decision Framing and Value-Focused Thinking

The initial phase of the project required a rigourous approach to decision framing to move beyond a narrow technical definition of the problem. Rather than viewing curtailment solely as a network balancing issue, the team adopted a Value-Focused Thinking (VFT) approach to identify the fundamental objectives of all stakeholders. This framing shifted the focus from "how to reduce curtailment" to "how to maximise the local value of renewable wealth for each stakeholder".

1 Discover Phase

Initially, the team used the 5Ws+1H Framework as a sense-checking tool to narrow the design brief and define the opportunity area. This initial analysis established the foundation for the curtailment reduction project goal.

WHAT?	WHO?	WHEN?	WHY?
What is the problem(s)? What information do you need to know? What type of data?	Who is your main stakeholder? Who are you solving this for? Who will use it/ be impacted by your solution?	When in the system is this needed? At what point?	Why is this intervention needed? Why is this problem/ set of problems happening?
Loss of energy from annual wind generation due to grid bottlenecks. Information needed: • generation data • residential demand patterns	Target both the 63% of Orkney households in fuel poverty, general households that use energy and the Network Operator (SSEN).	The intervention is needed in real-time during curtailment windows → periods where wind output exceeds the households demand	Curtailment is happening because local demand is currently rigid, misalligned with excess energy generation and insufficient to absorb the surplus wind power. This results in massive socioeconomic loss.

HOW?

How are you solving the problem? How is this linked to the stakeholder?
How is the process impacted by your solution?

By orchestrating consumption through smart appliances (automised household plugs, possibly charging EVs) and thermal storage (electric boilers/heaters) to create Demand Turn Up (DTU), aligning household use with wind surges.

HMW? (How might we)

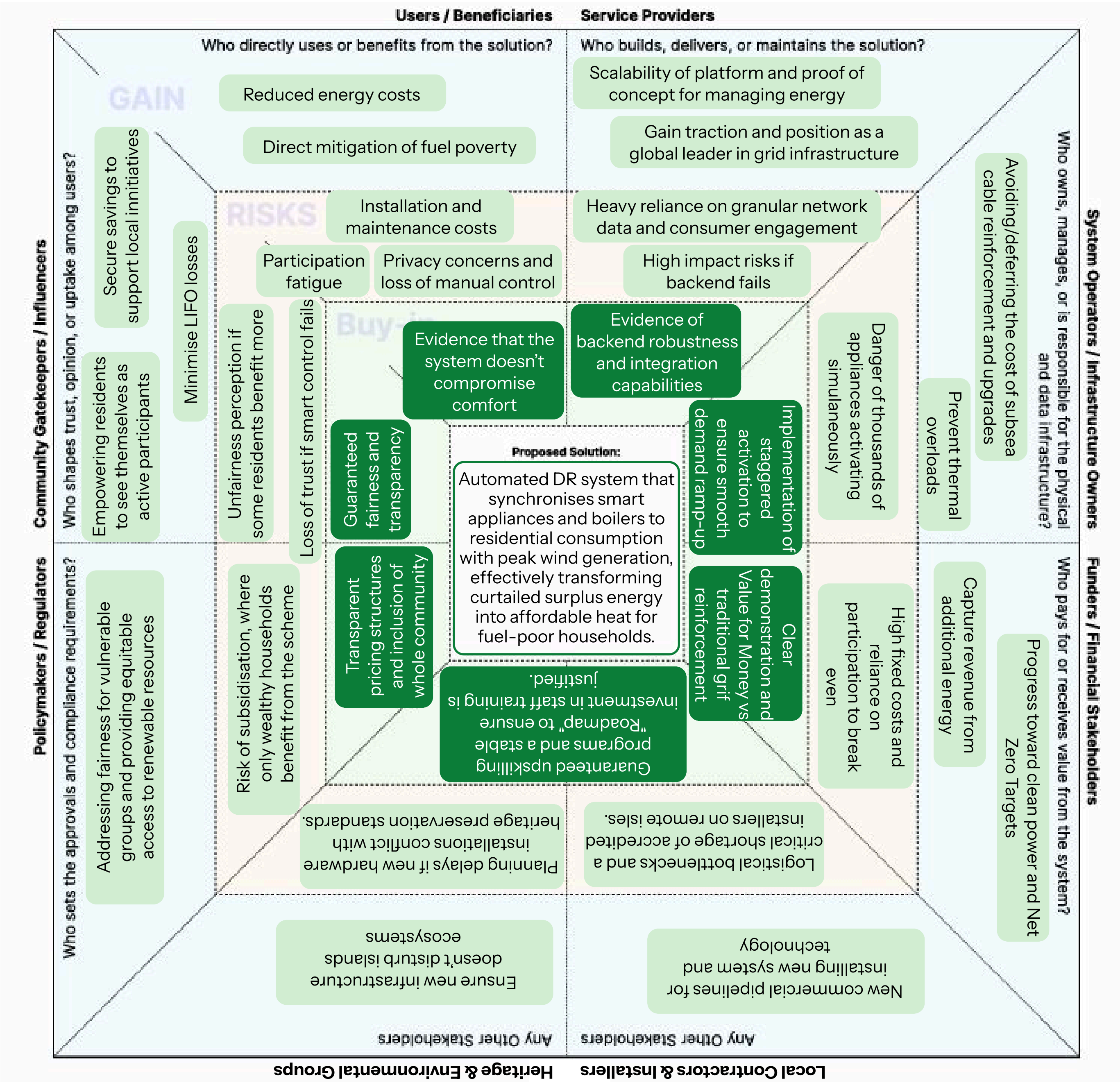
HMW is used to help innovators define their Brief clearly. Try it out, in one or two sentences: *How might we solve x with the use of y in order to...?*

How might we solve the renewable curtailment crisis with the use of smart demand-side response technologies in order to alleviate fuel poverty and defer expensive grid reinforcements?

Figure 1: *Sense-checking tool (5Ws +1H)*

2 Project Framing and the Stakeholder Universe

Following the discovery phase, the team used the Stakeholder Universe framework to critically review the proposed solution, identify gaps, and understand underlying tensions between stakeholders.



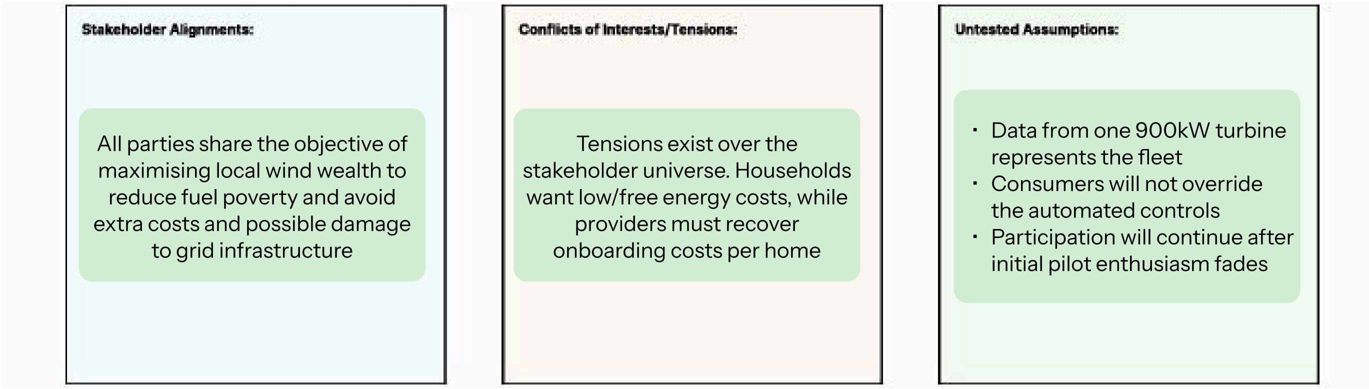


Figure 2: *Stakeholder Universe*

The primary insight derived from the stakeholder universe was that technical efficiency is meaningless without social support. By identifying Community Trusts as gatekeepers, we realised that the scheme's success hinges on procedural trust. Justifying a 20% community dividend was a strategic choice to align the goals of the individual consumer with the collective welfare of the isles, ensuring the project remains resilient even if individual participation fluctuates.

Descriptive Analytics – Quantifying the Addressable Market

Once the stakeholder landscape was defined, the analysis proceeded to quantify the exact volume of free energy currently rejected by the grid to provide the commercial justification for stakeholder investment. Using data from a representative 900 kW wind turbine and scaling it to the island-wide fleet, the team established a baseline for energy waste. Each group member initially analysed the dataset separately to explore different filtering methods, eventually converging to compare findings and contrast the resultant values.

Curtailment Modeling

The team selected the most conservative and defensible model with the following restrictions to ensure the business case remained credible under stakeholder scrutiny:

- To ensure the baseline represents actual wasted energy rather than low-wind periods, the team only counted intervals where actual power generation was within 1% of the restricted setpoint, filtering out maintenance downtime or low-wind scenarios, to only measure energy that was available but rejected by the grid.
- For every curtailed interval, the lost energy was calculated using the formula: *900kW - Setpoint x Duration.*
- The raw dataset covered approximately 2.63 years (23,022 hours). We averaged the total lost energy over this duration to create a representative 12-month baseline.
- A conservative financial value of £0.10 per kWh as a proxy for wholesale energy costs was applied.

Scale	Annual Energy Waste (Unrealised Power)	Annual Value Lost (Financial Opportunity)
Single Turbine (Rousay)	1,114,354 kWh (1.1 GWh)	£111,435
Orkney Fleet (23 Turbines)	25,630,144 kWh (25.6 GWh)	£2,563,014

Table 1: *Curtailment Analysis*

The decision to conduct independent analysis followed by a group convergence phase was a deliberate strategy to reduce groupthink bias and stress-test our assumptions. The primary insight found is that the grid is rejecting over £2.5M worth of clean energy annually, energy that is essentially free. This high potential opportunity justifies the high onboarding costs, as recouping even a portion of this loss provides a massive revenue boost for community generators while powering the DR network without requiring new generation assets.

Predictive & Behavioural Modeling

With the £2.5M annual opportunity quantified, the project transitioned into modeling the specific residential consumption patterns required to capture this wealth. The team transitioned from historical baselines to modeling potential behavioural shifts, using demand datasets for 5,430 households. This predictive phase analysed residential demand patterns against average curtailment events and seasonal behaviours.

Daily Consumption and Behavioural Constraints

The data revealed specific patterns in stakeholder behaviour that inform the system's technical constraints:

- A significant spike in demand occurs in the evening as people return home. Much of this is rigid energy that cannot be shifted without compromising quality of life.
- A large volume of unused flexible capacity exists overnight, ideal for automated smart appliances like washing machines.
- Modeled peak residential demand remains at ~5.04 MW, while curtailed surpluses often reach 15–20 MW.

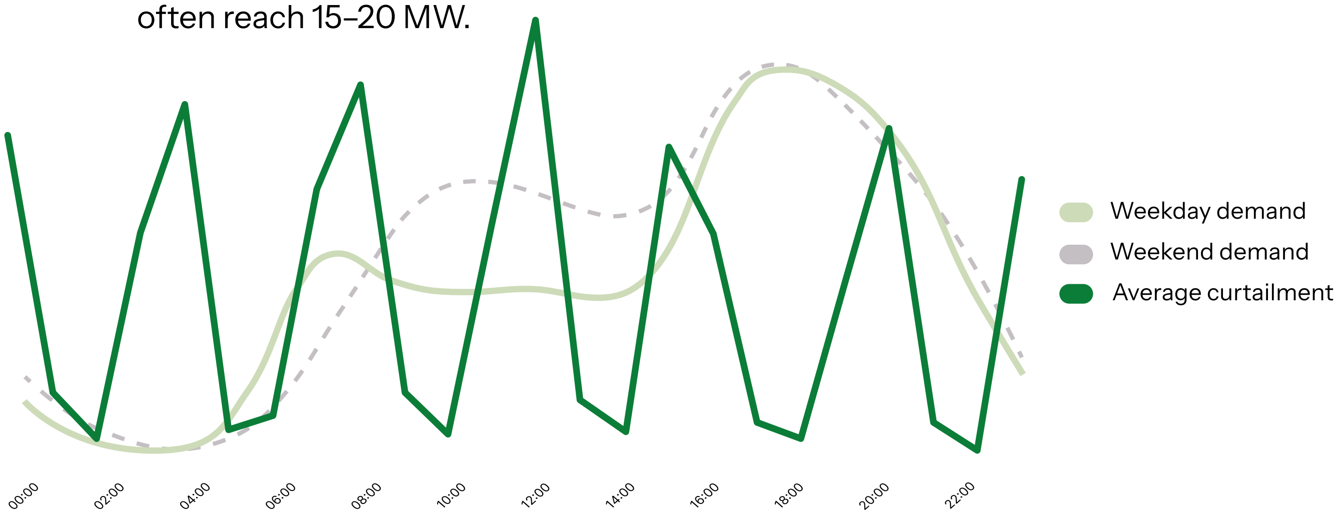


Figure 3: *Average Daily Energy Demand and Curtailment*

Solving Seasonality: The Two-Tiered Mitigation Strategy

Analysis of annual trends identified that curtailed energy spikes significantly between August and November. To address this, the team designed a two-tiered system to align variable supply with seasonal demand:

- Winter Mitigation: The project supports high-volume DTU through electric boilers, which absorb the massive curtailed surpluses during the high-heating-demand months (Jan–Mar and Oct–Dec). [4]
- Warmer Month Mitigation: During the warmer months when heating demand drops, the system uses smart plugs for appliances to mitigate the curtailed energy increase, ensuring the grid balancing service remains functional all year-round. [5]

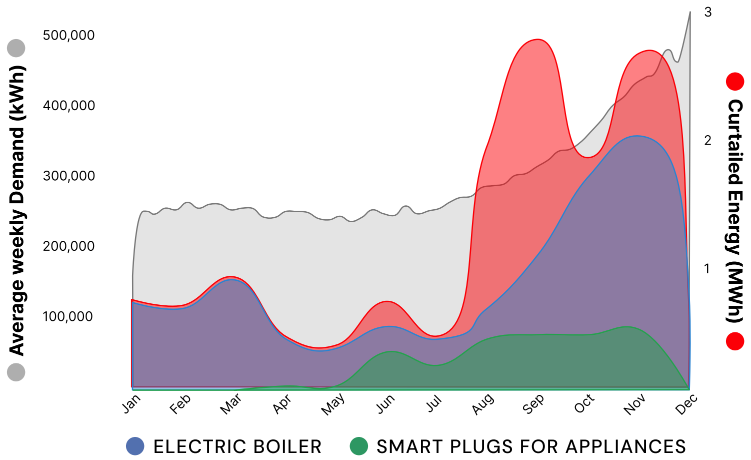


Figure 4: *Average Yearly Energy Demand and Curtailment*

The core insight was that manual human behaviour and seasonality are the primary bottlenecks for grid balancing. A heat-only solution would create a problem in summer, where thousands of pounds of hardware would sit idle while wind output continues to be curtailed. This justified a two-tiered approach: by integrating smart plugs for appliances, we ensure the project delivers value in the warmer months when boilers are less utilized.

Prescriptive Analytics & Incentive Design

Recognising that manual behavioural change and seasonality represent fundamental bottlenecks, the team translated these predictive insights into a pricing structure designed to secure participation and permission for automation. The primary shift involved moving from incentivising behaviour directly to incentivising permission for Kaluza to automate assets.

Stakeholder Value-Exchange Model

Each stakeholder pays into the system because they receive tangible value from the resulting flexibility.

Stakeholder	Contribution	Value Received
Network Operator	Installation costs (£90-£150/hh)	Grid stability; deferred costly reinforcement
Suppliers	Platform costs (£40/hh/year)	Revenue from otherwise wasted generation
Government	Fail-safe annual fixed overheads	Long-term community sustainability & equity
Households (hh)	Zero upfront costs; permission to automate control	Smarter homes; bill savings

Table 2: *Stakeholder Value-Exchange*

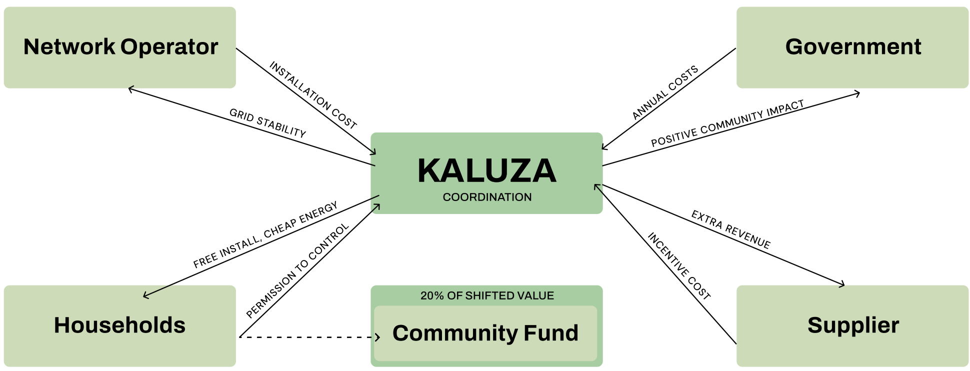


Figure 5: *Stakeholder Pricing Strategy*

Ensuring Fairness: The Community Dividend Pool

To address concerns that only households with large flexible loads would benefit, the team diverted 20% of total system savings into a community fund managed by the Orkney Community Trust.

This mechanism ensures the economic gains from flexibility are shared across the community. All participants receive equal voting power to allocate funds toward:

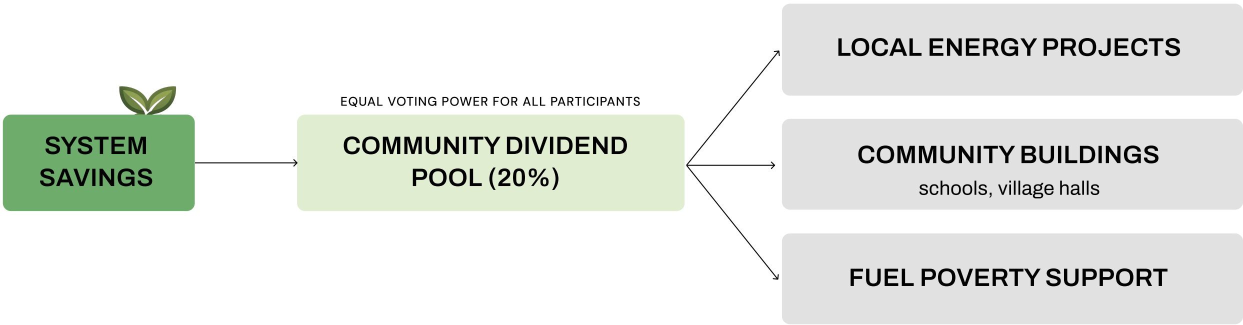


Figure 6: *Community Fund Allocation*

The team realised that the most significant barrier to adoption is not the technology, but the risk of creating a two-tier benefit system where low-flexibility households are left behind. By placing the community dividend at the center, we supported the project's fairness to the local council and regulator. The analytical insight here is that every participant, regardless of their flexible load size, has a stake in the island's renewable wealth. For the government, this is exactly what a desirable renewable scheme looks like: hitting Net Zero targets while ensuring happy consumers and visible social outcomes.

Risk Management & Uncertainty Analysis

Having justified the project's viability through a pricing model centered on fairness and energy citizenship, the analysis then turned to identify the operational dependencies and uncertainties that might still compromise the project. This resilience testing used dependency chain of risks and scenario planning to ensure the £2.5M recovery target.

Risk Dependencies

Three interconnected risks that must be managed in sequence were identified:

Risk	Likelihood	Impact	Mitigation
Regulatory (Initial Risk)	Low	High	Involve consumer protection early in the incentive design.
Participation (The Backbone)	Medium	Medium	Define a minimum viable household threshold before launch, supported by a backup recruitment pool.
Technical (Operational Proof)	Medium	High	Build redundant communication pathways and staggered activation algorithms.

Table 3: *Risk Dependencies & Mitigation Strategies*

Major Uncertainties and Scenario Targets

The team identified three primary uncertainties: energy use habits (override for comfort), shifting funding landscapes (reliance on grants), and external environmental factors (low-wind years vs. major storm spikes).

To ensure financial resilience, the project models three plausible futures :

- Probable Case: 3,000 HH (Average mix of appliances).
- Best Case: 1,800 HH (Prioritizing high-flexibility assets).
- Worst Case: 5,500 HH. Strategic Response: If engagement is lower than expected or prices drop, we simply expand by onboarding more households to meet the sink capacity.

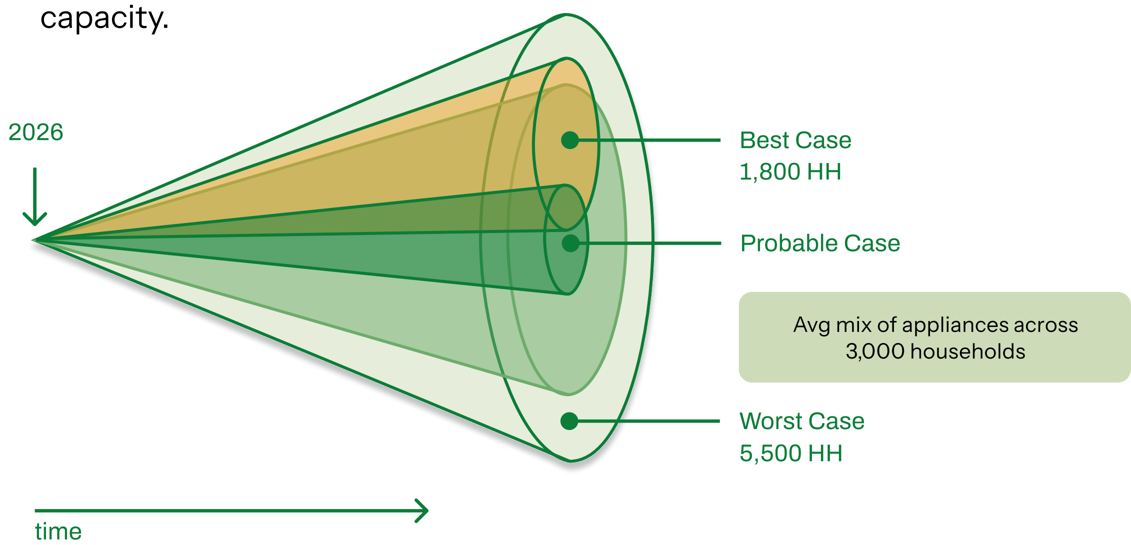


Figure 7: *Scenario Planning*

The team reasoned that standard project management relies on static forecasts, which are dangerous in a climate-dependent system. Scenario planning was justified because it allows the project to model a viable path even for the worst case. The analytical insight here is that social risk cannot be fixed, only accommodated. By planning for automated overrides and a scaling strategy in the worst case, we justified a project that is flexible and adaptive, ensuring the £2.5M revenue recovery.

Critical Reflection – Energy Justice and Trade-offs

Following the development of an adaptive roadmap, the final phase of the analysis evaluated the proposal's standing to ensure the project delivered on its promise of fair transition.

- Distributional Justice: Identifying the possibility where only wealthy households with high flexibility assests benefit and mitigating the digital divide by ensuring low-income households benefit first.
- Procedural Justice: Using the community fund to give all residents an equal vote in how recovered wealth is spent, ensuring inclusivity regardless of a home's flexibility potential.

The team's reflection concluded that the most difficult trade-off was between automation and education. While Kaluza's automation ensures high reliability, it limits the energy literacy gained through manual habit change. However, we justified the automation-first approach due to the urgency of fuel poverty. Using the community dividend to provide a visible, shared outcome that recognizes even the most vulnerable residents.

Conclusion

The HSO analysis demonstrates that Demand-Side Response is a robust solution to the Orkney landscape. By moving from initial framing to prescriptive pricing, the team created a framework that balances the operator's need for stability with the community's need for energy. The final recommendation is a 3-phased roll-out:

- Phase 1 → Social Heat Anchor
- Phase 2 → expanding to private Wind-Followers
- Phase 3 → trading flexibility as a national asset

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